Resource Allocation in Beam Hopping Communication System

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***Abstract*—Beam-Hopping provides satellite systems with an additional dimension of agility making it possible to adjust the allocation of the scarce satellite spectral and power resources to the demand which varies in space and geography as well as with time. Provided such flexibility, it has a great potential in enhancing communication satellite capacity. An effective resource allocation mechanism which can meets future demands flexibly is urgently needed in order to achieve the best user communication quality and to provide more users with more satisfactory service. The existing beam steering mainly focus on covering the beam position which largest demand the spectrum resources and do not fully utilize the advantages brought by the flexibility in beam range. In this paper, we propose a beam resource management algorithm that adjusts the beam size according to the service distribution, which can make more users accessible to make full use of beam bandwidth resources by covering more beam positions. Simulation results show that the proposed algorithm can effectively reduce packet loss rate, improve the quality of service (QoS), and guarantee service priority.**

***Index Terms*—beam hopping, RRM, system capacity, QoS**

I. Introduction

Maximization of communication resources utilization is the core issue of communications satellite resource management.The algorithm of the stationary multi-beam resource management based on the Multi-Frequency Time-Division (MF-TDMA) system is quite mature, and the bandwidth resource allocation can be dynamically adjusted according to the user service requirements, so that the system capacity can be effectively improved when given bandwidth resources.Beam-Hopping has great advantages over traditional multi-beam antennas in terms of combating a strong directional interference, flexibility, reducing power consumption, and increasing system capacity.

The Beam-Hopping mechanism has become a hot research topic cause of its outstanding advantage in combating a strong directional interference and flexibility, which can effectively increase the capacity of the satellite communication system. In [1], illustrated the merits of Beam Hopping Techniques by analyzing the physical structure and assessed the power consumption and flexibility by comparing the hopped and non-hopped systemsand also introduced power controlling method using genetic algorithm which is based on the assigned capacity requirements. Considering the interference between the beams, [2] analyzes the performance advantages of beam hopping in combating a strong directional interference.Reference [3] proposes two heuristic algorithms to maximizes system capacity which subject to maximizing signal to interference plus noise ratio (SINR) and minimizing Co-Channel Interference (CCI) respectively. According to the load structure of the Beam Hopping, [4] proposes a genetic algorithm to balance system capacity and power consumption. All these algorithms have high signal-to-noise ratio and large system capacity as the optimization goal, and no specific service allocation strategy is given. In Beam Hopping mechanism, the core of resource management is to allocate bandwidth resources effectively taking advantage of flexibility to achieve maximum users' satisfaction. [5] proposes a organized beam-hopping techniques to effectively improve the user's performance at the boundary between two adjacent wave positions. [6] designed Beam Hopping pattern subjecting to long-term delay fairness, which has improved in fairness. However, the specific beam hopping illumination strategy has not given yet.

The Beam-Hopping is currently used in satellite communications. In the spaceway3 communication satellite launched in August 2007, the downlink spot beam adopts the TDM high-speed multiplexing mechanism, in which a different subset of all the beams is illuminated according to the service arrival intensity and destination address such that a single beam can offered as much capacity as the service required. The downlink port support carrier rate can be changed dynamically according to service requirements.

Considering that a single wave position can not make full use of the total spectrum resource of the beam sometimes, an algorithm that can adjust the illumination position and coverage with the service distribution is given. The highlight of it is to keep balance between the increase of resource utilization due to the increased coverage and the reduce of total resource, so as to improve system capacity and user satisfaction.

The rest of this paper is organized as follows. Section II introduces the beam hopping scenario and provides the system assumption. A algorithm to solve the problem is proposed in Section III. Simulation results are presented in Section IV and conclusions are drawn in Section V.

II. System Assumptions

The bandwidth resources on the satellite are extremely valuable. The beam hopping technology can be used for

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efficient communication between the satellite and the ground terminal, which greatly improves the bandwidth utilization and make sure a better communication quality of ground users. The application scenario of the beam hopping is designed in this section and also analysis the variation features of beam hopping.

A. Scenario Design

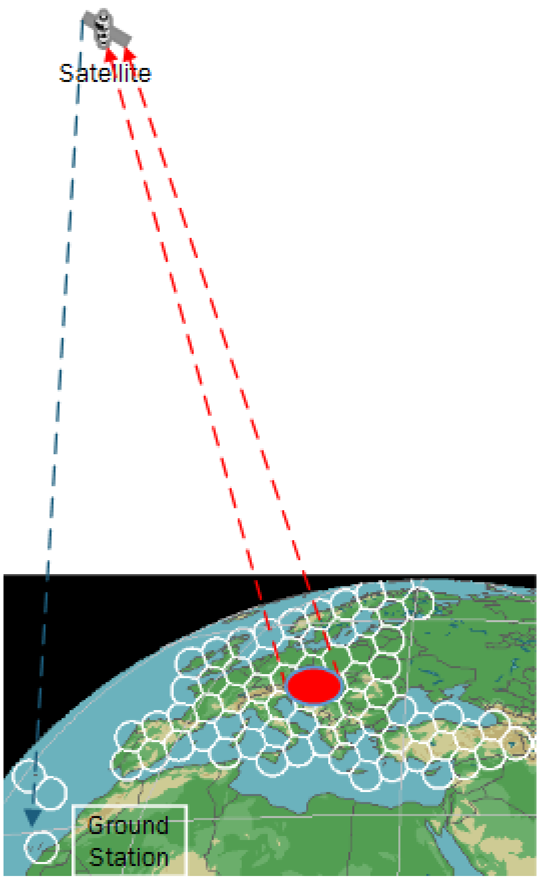

Fig. 1. scenario design

In the scenario, the ground terminal transmits a resource application frame to a satellite node through a global beam at the beginning of business origination. The satellite allocates beam resources according to requirements and delivers the resource allocation result through the global beam. After the ground terminal is approved, it transferred to the beam and transmit the service data in the allocated uplink carrier and time slot. After receiving the service data, the satellite directly sends the service data to the ground station node. We consider the packet is successfully transmitted only when the ground station receives it. The user who does not receive the resource have to wait for the next allocation period. Key features of the system is assumed as

- Beam layout is geographically distributed into 34 positions in the region of China. The neighbor relations are presented in Fig.2.

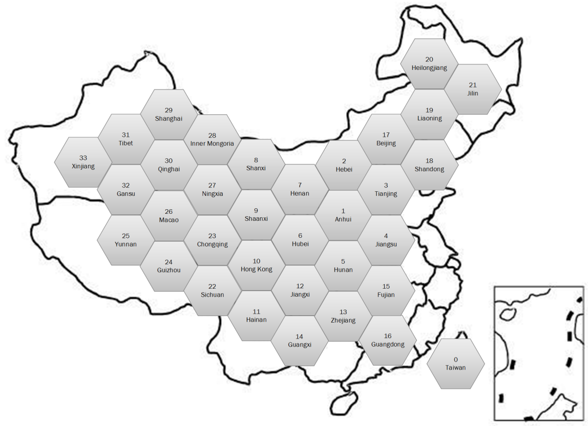

Fig. 2. beam layout

- There are three beams on the satellite, each single beam has a capacity of 500M, and the beam is hopping independently. The range can be changed dynamically and multiple users can share channel resources by using multiple access methods. The satellite-based resource manager allocates suitable resources for the user based on the service application sent by the user.
- The user is evenly distributed among the 34 beam positions.We choose constant bit rate (CBR) services of a certain size which is continuously transmitted at certain intervals between the start and end times and the termination time. The services are randomly initiated and randomly terminated.
- The resource application frame format is

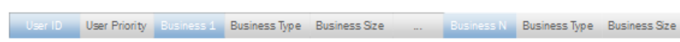

Fig. 3. frame structure

The resource allocation algorithm for the satellite resource manager is given in the algorithm section.

B. Communication Capabilities

1) Beam resources and coverage changes

Assuming that the bandwidth

$$BW = f(R) \tag{1}$$

R is beam coverage radius, the bandwidth decreases with increasing radius R. Considering the cost of beam adjustment and reducing the computational complexity, the initial value of radius is 1 and it can be changed step according to the service distribution, the maximum of R is 4. The following table depicts the variation of the beam bandwidth with the radius of the beam coverage is as follows. R = 1 indicates that the beam covers only the current single beam position, R = 2 indicates that the beam covers a central wave position and its adjacent 6 beam positions and so on. Due to the decrease of the beam capacity and the interference of spatial noise when the beam range is expanded, the bandwidth of a single beam will be less than 1/K of the total bandwidth when the beam coverage becomes K beam positions. The beam bandwidth capability as the radius changes listed in TABLE 1.

TABLE I
BEAM RESOURCES VS COVERAGE RADIUS

R	**1**	**2**	**3**	**4**
BW = f(R)	f(1)	f(1)/8*7	f(1)/21*19	f(1)/40*37

2) Terminal capability and beam range/converage
   The ground terminal equipment has various sizes, weights, power consumption, and applications. The antenna size and communication frequency band are also different. For simplicity, the system classifies it into four levels of A, B,C,D. The relationship between communication capability and beam coverage is

TABLE II
TERMINAL CAPABILITY VS COVERAGE RADIUS

cat \ **R**	1	2	3	4
A	✓	✓	✓	✓
B	✓	✓	✓	×
C	✓	✓	×	×
D	✓	×	×	×

In this table,"✓"indicates that the terminal can communicate with the satellite under the coverage of the beam, and "×" vice versa. When the beam coverage radius is 1, all terminals can communicate with the satellite. When the radius of the beam expands to 2, the user whose capability is D cannot communicate with satellite. In this scenario, the terminal capability is evenly distributed to each user.

III. Algorithm

A. Algorithm Idea

Assuming that the beam covers one beam position, the bandwidth capability is 300 M. According to the corresponding relationship between the beam bandwidth and the coverage radius changes in the previous system model, when R=2, the bandwidth capability is reduced to 300/(8*7)=262.5 M. When the traffic distribution is as shown in the above figure, the resource utilization rate is only 90/300 if the beam covers only the 1st wave position, and only 90M traffic can be guaranteed. In this case, the increase of the beam radius to 2 can be considered. The beam capacity reduce to 262.5M, the total traffic volume of all wave positions is 300M which is greater than the total capacity of the beam. Therefore, the maximum amount of traffic is 262.5M. At this time, the bandwidth utilization rate is 1, and the service traffic is 262.5M. The performance is significantly improved compared to the mechanism that covers only one beam position.

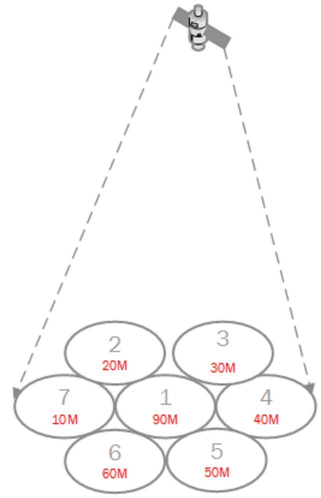

Fig. 4. spot beam capability

However, terminal capabilities need to be considered in reality. When the beam capacity is degraded, the weak terminal will lose communication capabilities, the problem is transformed into the bandwidth and the carrier capacity changes, terminal capabilities, according to the beam coverage in the case of uncertain business distribution. Solve the beam coverage position and radius with the goal of optimizing resource utilization.

But in reality, terminal capabilities need to be considered. When the capacity of the beam drops, the weak terminal will lose its communication capability. The problem will be converted into obtain the beam position and beam coverage which should be illuminated by hopping beams when the service distribution is uncertain. Resource utilization is the optimization goal here and the restrictions are bandwidth due to beam coverage, the change of the carrier capacity and the terminal capability.

$$\begin{cases} \text{1.business is lunched randomly;} \\ \text{2.carried capability} \geq \text{terminal capability;} \\ \text{3.beam resource BW = f(R)} \end{cases}$$

η = the total traffic in the current beam coverage / the band width of one spot beam

s.t. get a good solution of coverage area and radius when η is maximum in current scenario and constraint.

B. Algorithm Flow

This algorithm aims to access more services by expanding the beam illumination range when the resource utilization is below a certain threshold, and to increase the system resource utilization to meet more user needs. The total single spot beam bandwidth is f(1) .When the beam irradiation radius is R, the beam bandwidth is f(R) , and the maximum bandwidth utilization rate at this time is f(R)/f(1) , so the expansion of the beam range inevitably brings reduction in the total bandwidth resources. Therefore, the beam range would not expand when the maximum area traffic volume buss$_{max}$(R) under the coverage radius R is bigger than the bandwidth f(R+1) under the coverage radius R+1. Fig.5 shows the specific process of the algorithm.

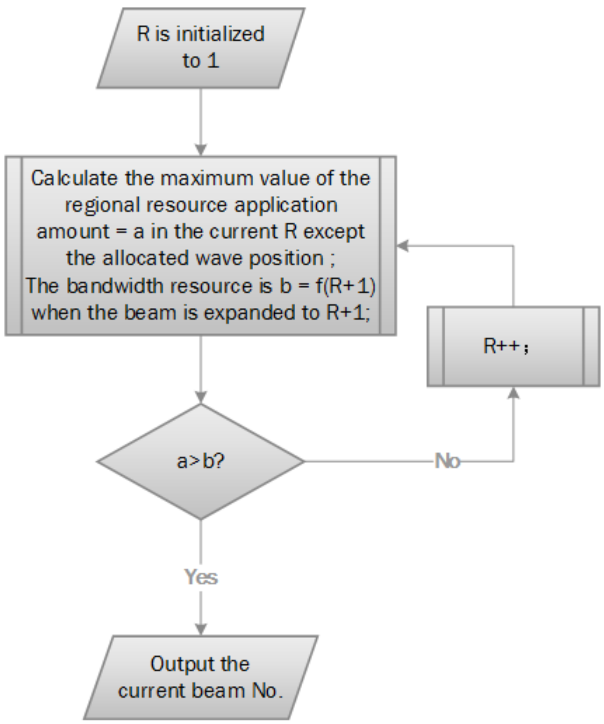

Fig. 5. flow chart

The initial value of the beam coverage radius is 1. When the resource utilization rate is less than the maximum resource utilization rate after the beam is expanded, the beam radius increases, and then the resource utilization under the new radius is recalculated; otherwise, the beam radius is determined and the beam position number whose business applications amount is largest would output. The beam coverage range where the beam can support the minimum terminal corresponding carrier capability is the limit. When the radius is greater than it, all terminals cannot communicate with satellite normally, which is meaningless. The on-board resource manager obtain the beam position group which with the highest resource utilization and record it through the above process. If the current number of beams is N, the resource manager needs to perform the above operation within a super frame period to obtain beam position groups in the corresponding coverage area. To avoid double calculations and inter-beam interference, the new calculation does not take the output before into consideration. After this, the next question to be considered is how to effectively access the user under the beam coverage, so as to allocate a proper amount of frequency resources to the user in a fair and efficient manner. We take service priority as the allocation order. When the remaining fragments are insufficient to meet the demand for low-priority traffic, the service would be accessed which with less traffic than the left fragment size and higher priority.

IV. Simulation and Analysis

The algorithm proposed in section III is to determine the hopping strategy and beam size according to the service distribution. In order to test the effectiveness of the algorithm when the system model of section II is considered, a scenario was constructed in a discrete simulation system. The packet success rates of the two methods, variable and fixed beam range, are given respectively. Evaluation index focuses on the packet loss rate and high-priority users' service satisfaction rate.

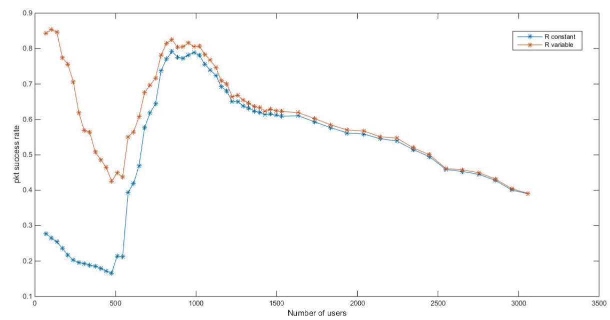

Fig. 6. total packet success rate versus the number of users

The performance of total packet success rate with respect to the number of users is shown in Fig.6. Obviously, the more users, the more packets are generated. The blue curve in the figure indicates the success rate with the increase of the number of users under the fixed beam range mechanism. The yellow curve indicates that the mechanism whose beam range is variable could increase the success rate when the resource utilization rate is low. When the abscissas are the same, the business distribution of the two curves is exactly the same, so it can accurately explain the difference between the two mechanisms. The results show that the mechanism whose coverage is variable according to the service distribution has significant performance in reducing the packet loss rate. With the increase of the number of users, the gap between the beam expansion and the non-expansion of the beam loss is smaller and smaller, that is because when the service of a single beam position already can fully use the total resource capacity of a single beam.In this case, the beam cover only one beam position to utilizes the total bandwidth resources effectively, avoiding system performance plummeted cause of the total resources decrease due to beam expansion. In fact, although the gap between the two approaches is decreasing with the business increasing, the number of lost packets due to the small packet loss rate is very large since the base number of packets is already very large due to a large amount of traffic. Therefore, the beam expansion method still has advantages in improving system capacity.

In order to show that the impact of increasing the total resource utilization caused by beam expansion increases on the services' priority,the relationship between the service success rate and the priority is analyzed when both methods reach the maximum resource utilization. Fig.7 draws the simulation results.

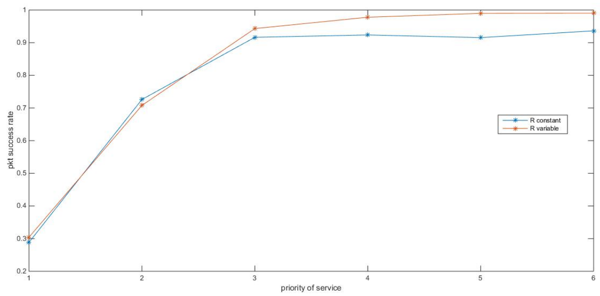

Fig. 7. total packet success rate versus services priority

Fig.7 illustrates that services with high priority under the two mechanisms show lower packet loss rates. The method of expanding the beam range would increase the success rate of the total services while still ensuring a higher success rate for high-priority services. As more high-priority users are accessed, the packet loss rate of high-priority services is effectively reduced compared to a fixed beam range where the packet loss rate of low-priority services is comparable under both mechanisms.

V. Conclusion

A algorithm is proposed in this paper to avoid wasting bandwidth sources when services are distributed broadly and small. The simulation results also show its good nature to the services with high priority. High-priority services have greater probability to obtain resources, so the packet loss is reduced. This method of beam-range expansion is significant to maximize the effectiveness of application layer users when the amount of traffic is not large. With the increase of business volume, the beam covers only one beam position already can fully utilize the total single-point beam capacity since the amount of resources for a single wave position is already very large. It is not so necessary to expand the beam range in this case. Because the beam expansion enables more services to access, this algorithm also improves QoS in terms of fairness. The beam hopping has obvious advantages in anti-jamming, anti-interception, and system performance enhancement, but the text does not consider the interference that may be caused between the beams.The power of each antenna can also be adjusted to the change of the channel conditions in the future work, thereby correcting the signal to noise ratio declines caused by factors such as rain attenuation, etc.

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